LAB 8 : Server Configuration – DHCP, DNS and Web Server in Cisco Packet Tracer

OBJECTIVES :

- To configure a server to provide DHCP, DNS, and Web (HTTP) services in Cisco Packet Tracer.
- To understand how DHCP assigns IP addresses and how DNS resolves domain names in a network.
- To verify the proper functioning of server services by accessing the web server from client devices.

THEORY :

1) **Server Configuration :**

   Server configuration refers to setting up a server so it can deliver network services to client devices. Typically, a server is assigned a static IP address to ensure that clients can consistently access its resources. In this lab, one server was configured to provide DHCP, DNS, and Web (HTTP) services. Proper configuration enables reliable communication, centralized service management, and efficient operation of the overall network.

2) **DHCP (Dynamic Host Configuration Protocol) :**

   DHCP (Dynamic Host Configuration Protocol) is a network protocol that automatically assigns IP addresses and related network settings—such as subnet mask, default gateway, and DNS server—to client devices. Rather than configuring each computer manually, DHCP provides these parameters dynamically when a device connects to the network. This approach minimizes configuration errors, saves administrative time, and simplifies overall network management. In this lab, the DHCP server was configured with an IP address pool, which it used to automatically allocate addresses to client PCs.

3) **DNS (Domain Name System) :**

DNS (Domain Name System) translates human-readable domain names into IP addresses. It is much easier to remember a name such as **www.hcoe.com** than a numerical IP address. The DNS server maintains records that map each domain name to its corresponding IP address.

When a client enters a domain name to access a website, the DNS server processes the request, resolves the name, and returns the appropriate IP address. In this lab, DNS records were configured so that clients could reach the web server using its domain name instead of typing its IP address directly.

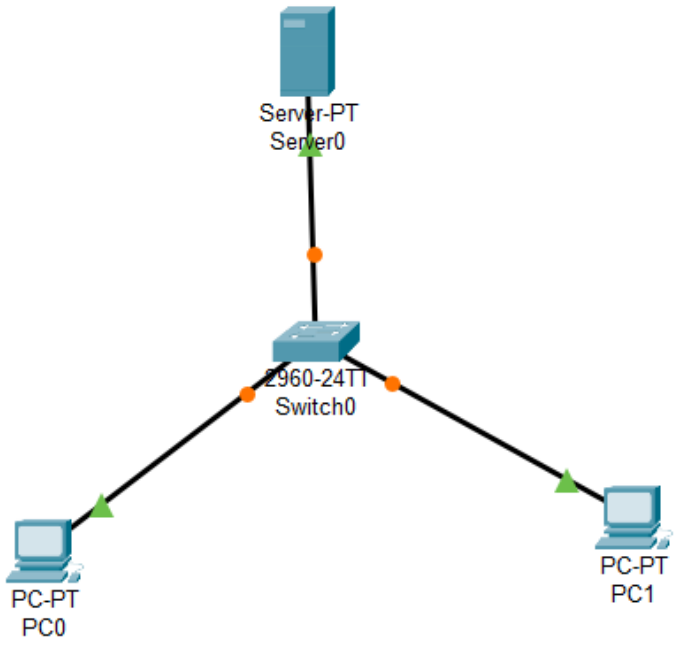
4) **Web Server (HTTP Service) :**

A web server provides web pages to clients using the HTTP protocol. It hosts website files such as HTML pages and responds to client requests made through a web browser. In this lab, the HTTP service is enabled on the server, allowing client PCs to access a webpage using a browser. This demonstrates how websites are hosted and accessed within a network environment.

Integration of DHCP, DNS, and Web Server :

DHCP, DNS, and Web Server services function together to create a fully operational network environment. DHCP automatically provides IP addresses and essential network settings to client devices. DNS translates domain names into IP addresses, allowing users to access resources using easy-to-remember names. The web server then delivers the requested web content to the client.

Network Topology :



Configuration :

IP Addressing Scheme

Device	IP Address	Subnet Mask	Gateway
Server	192.168.1.2	255.255.255.0	192.168.1.1
PC	DHCP	DHCP	DHCP

For PCs,

Device	IPv4 Address	Subnet Mask	Default Gateway	DNS Server
PC0	192.168.1.1	255.255.255.0	0.0.0.0	192.168.1.2
PC1	192.168.1.3	255.255.255.0	0.0.0.0	192.168.1.2

For Server,

Device	Default Gateway	DNS Server
Server0	192.168.1.1	192.168.1.2

Procedure :

Server Configuration :

Step 1 : Assign Static IP to Server

1. Click on Server
2. Go to Desktop → IP Configuration
3. Configure the following:

- IP Address: 192.168.1.2
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.1.1
- DNS Server: 192.168.1.2

DHCP Configuration :

Step 2 : Enable DHCP Service

1. Go to Server → Services → DHCP
2. Turn DHCP: ON

Step 3 : Create DHCP Pool

Field	Value
Pool Name	LAB_POOL
Default Gateway	192.168.1.1
DNS Server	192.168.1.2
Start IP Address	192.168.1.10
Subnet Mask	255.255.255.0
Maximum Users	50

Click Add **Verification**

:

On each PC:

- Go to Desktop → IP Configuration
- Select DHCP

```
C:\>ping 192.168.1.3

Pinging 192.168.1.3 with 32 bytes of data:

Reply from 192.168.1.3: bytes=32 time=2ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 2ms, Average = 0ms
```

DNS Server Configuration :

Step 4 : Enable DNS Service

1. Go to Server → Services → DNS
2. Turn DNS: ON

Step 5 : Add DNS Records

- Name: www.hcoe.com
- Type: A record
- Address: 192.168.1.2

Click Add **Verification**

:

On each PC:

- Open Command Prompt
- Ping: www.hcoe.com

```
C:\>ping 192.168.1.3

Pinging 192.168.1.3 with 32 bytes of data:

Reply from 192.168.1.3: bytes=32 time=2ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 2ms, Average = 0ms

C:\>ping www.hcoe.com

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

```
C:\>ping www.hcoe.com

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time=13ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 13ms, Average = 3ms
```

Web Server Configuration :

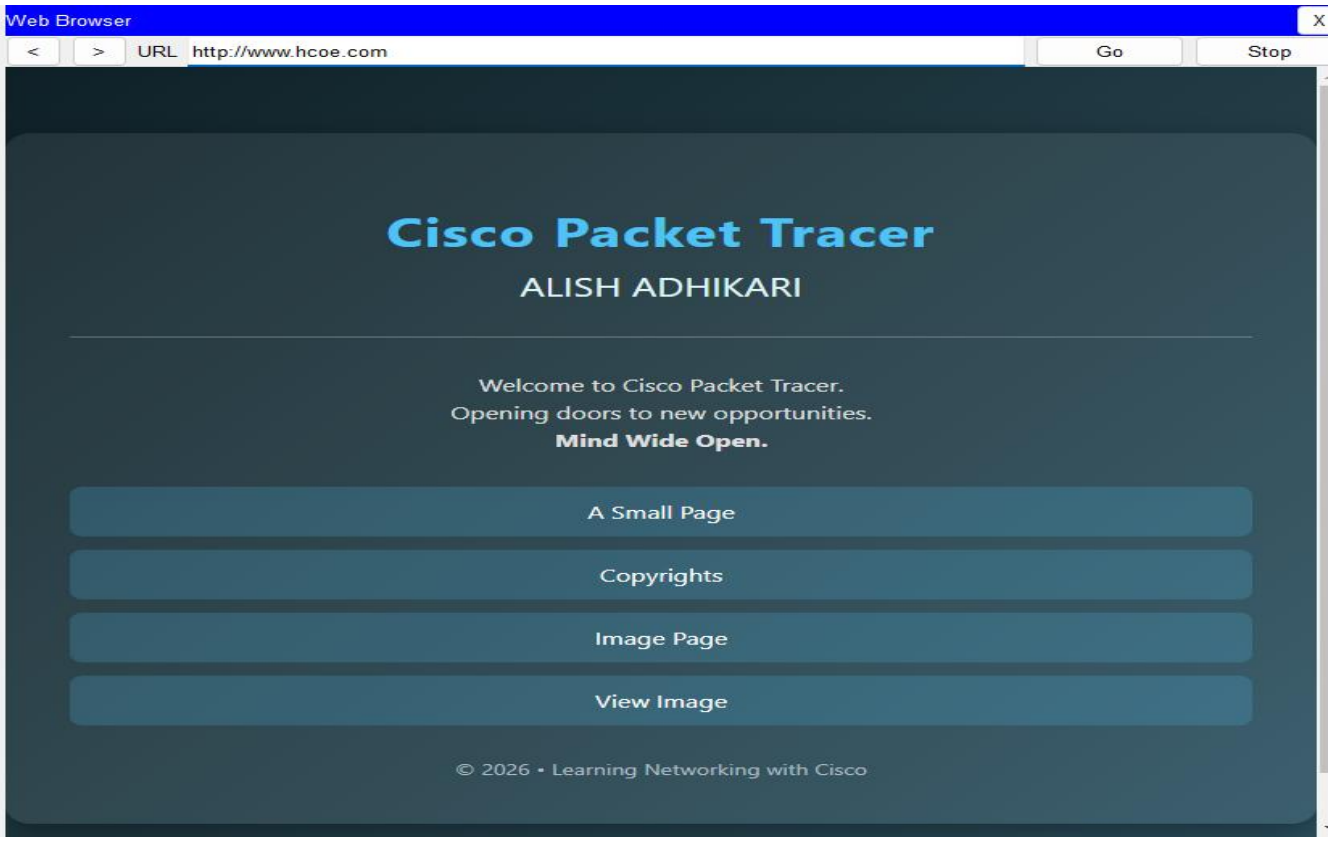
Step 6 : Enable HTTP Service

1. Go to Server → Services → HTTP
2. Turn HTTP: ON
3. Edit index.html (optional)

Verification :

On PC:

- Go to Desktop → Browser
- Type: http://www.hcoe.com



RESULT :

In Cisco Packet Tracer, the DHCP, DNS, and web server were configured successfully according to the lab objectives. The server received a static IP address, and all client PCs automatically obtained valid network settings via DHCP. DNS name resolution functioned correctly, enabling clients to access the web server using the configured domain name. The HTTP service was activated, and the webpage was successfully accessed from client devices, confirming that all services were operating properly.

DISCUSSION :

This lab illustrated the practical implementation of key network services and their importance in enabling efficient communication. DHCP automated IP address assignment, reducing manual configuration, while DNS facilitated easy access to network resources via domain names. Successful access to the web server confirmed proper application-layer communication.

Additionally, the lab enhanced understanding of service dependencies, correct configuration of parameters, and basic troubleshooting within a simulated network environment.

CONCLUSION :

The objectives of configuring DHCP, DNS, and HTTP services on a server in Cisco Packet Tracer were successfully achieved, with proper IP address assignment, domain name resolution, and verified web server access from client devices.